Introduction

Landmark analysis is the standard fix for the immortal-time bias that arises when subjects are classified by a post-baseline exposure (treatment response, biomarker change, secondary event). Naively grouping subjects by a covariate measured after follow-up begins assigns "responder" status only to subjects who survived long enough to be assessed, automatically biasing comparisons in their favour. Landmark analysis sidesteps this by selecting a fixed landmark time, restricting to subjects alive and observable at that time, classifying them by covariate value *at the landmark*, and then analysing only the post-landmark survival.

Prerequisites

A working understanding of Cox regression, time-varying covariates, and the immortal-time bias that makes naive responder-vs-nonresponder analyses invalid.

Theory

At a chosen landmark time L:

1. Exclude subjects with an event before L — they cannot be classified by status at L.
2. Classify each remaining subject by their covariate value at L.
3. Treat L as the new time origin and analyse the residual survival $T_i - L$ for $T_i > L$.
4. Fit a standard Cox model on the post-landmark data.

Multiple landmarks can be combined into a "super model" with `id`-clustered variance and a smooth function of L as a covariate, gaining efficiency at the cost of additional model complexity (Van Houwelingen 2007).

Assumptions

The covariate value at landmark L is the relevant exposure for prognosis; sufficient follow-up beyond L remains; and the censoring mechanism after L is non-informative.

R Implementation

```r
library(survival)

set.seed(2026)
n <- 200
d <- data.frame(
  id = 1:n,
  time = rexp(n, 0.1),
  status = rbinom(n, 1, 0.6),
  response = rbinom(n, 1, 0.5)      # binary time-varying covariate
)

L <- 6   # landmark at 6 months

# Subjects at risk at L
at_risk <- d[d$time >= L, ]
at_risk$time_from_L <- at_risk$time - L
at_risk$status_from_L <- at_risk$status

fit <- coxph(Surv(time_from_L, status_from_L) ~ response, data = at_risk)
summary(fit)
```

Output & Results

A standard Cox regression on the post-landmark cohort: hazard ratio for the covariate (here `response`) measured at the landmark, with 95 % confidence interval and Wald test. The number of events and at-risk subjects shrinks after the landmark, so confidence intervals are wider than in a full-cohort analysis.

Interpretation

A reporting sentence: "Among the 152 patients alive at the six-month landmark, treatment responders (n = 78) had a 50 % lower hazard of subsequent death than non-responders (HR 0.50, 95 % CI 0.30 to 0.83); the analysis avoids the immortal-time bias inherent in unrestricted responder-vs-non-responder comparisons." Always state the landmark time and the post-landmark sample size.

Practical Tips

- Choose the landmark time at a clinically meaningful assessment point (six-week imaging response, three-month biomarker re-measurement); arbitrary times can produce unstable estimates.
- Multiple landmarks (super-model approach) combine information across time and provide a smooth treatment-effect estimate; cluster the variance on subject id when fitting.
- Sample size drops at the landmark; power calculations should account for the post-landmark cohort, not the original.
- Compare landmark results with a full time-varying Cox model that uses the covariate as a `tt()` term; both should agree if the landmark choice is reasonable.
- Be transparent about exclusions: report how many subjects had events before L and were excluded, because this is the headline difference from a naive analysis.
- For complex time-varying patterns, consider joint longitudinal-survival models (`JM` or `joineRML`) instead of repeated landmark analyses.

R Packages Used

`survival` for the standard Cox fit on landmark cohorts; `dynpred` for super-model landmark analyses with smooth time effects; `JM` and `joineRML` for joint longitudinal-survival modelling that handles continuous time-varying covariates without landmarks; `mstate` for multi-state extensions.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Survival primer: KM, log-rank, Cox PH
- Time-varying covariates and immortal-time bias
- Competing risks and multistate models
- Survival ML (random survival forest, DeepSurv)
- Time-dependent Brier, IPA, external validation